MATH 345 Skeleton Notes
Unit 1: Vectors, matrices, and linear maps
Section 1.3: Matrix-vector product and linear maps

Vectors as columns

A vector can be written as a row or as a column. When a matrix acts on a vector in this course, we usually write the vector as a column:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}.$$

From now on, when we write $\mathbf{x} = (x_1, \dots, x_n)$, $\mathbf{x}$ is interpreted as a column vector.
If A is an $m \times n$ matrix, then A has n columns. We write

$$A = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{bmatrix}$$

where each $\mathbf{a}_j$ is a column vector in $\mathbb{R}^m$.

Matrix-vector product

Definition: Matrix-vector product

Let $A = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{bmatrix}$ be an $m \times n$ matrix and let $\mathbf{x} = (x_1, \dots, x_n)$ be a column vector in $\mathbb{R}^n$. The

$\boxed{\text{matrix-vector product}}$ is

$$A\mathbf{x} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n.$$

Activity: Computing a matrix-vector product

Compute $A\mathbf{x}$ where

$$A = \begin{bmatrix} -1 & 4 & -5 \\ 3 & 1 & -2 \end{bmatrix}, \qquad \mathbf{x} = \begin{bmatrix} 2 \\ -3 \\ 4 \end{bmatrix}.$$

Using columns,

$$A\mathbf{x} = 2\begin{bmatrix} -1 \\ 3 \end{bmatrix} - 3\begin{bmatrix} 4 \\ 1 \end{bmatrix} + 4\begin{bmatrix} -5 \\ -2 \end{bmatrix}$$
$$= \begin{bmatrix} -34 \\ -5 \end{bmatrix}.$$

Theorem: Properties of matrix-vector products

Let A and B be $m \times n$ matrices, let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, and let r be a scalar. Then

- $A(\mathbf{x} + \mathbf{y}) = A\mathbf{x} + A\mathbf{y}$,
- $A(r\mathbf{x}) = r(A\mathbf{x}) = (rA)\mathbf{x}$,
- $(A + B)\mathbf{x} = A\mathbf{x} + B\mathbf{x}$.

Theorem: Row dot-product view

If A is an $m \times n$ matrix with rows $\mathrm{row}_1(A), \ldots, \mathrm{row}_m(A)$, then

$$A\mathbf{x} = \begin{bmatrix} \mathrm{row}_1(A) \cdot \mathbf{x} \\ \mathrm{row}_2(A) \cdot \mathbf{x} \\ \vdots \\ \mathrm{row}_m(A) \cdot \mathbf{x} \end{bmatrix}.$$

Note: Rows measure; columns contribute

A matrix-vector product has two complementary readings. In the row view, each row measures the input by taking a dot product with $\mathbf{x}$. In the column view,

$$A\mathbf{x} = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n,$$

so the input coordinates tell how much each column contributes to the output.

Examples of matrix-vector products

The same product $A\mathbf{x}$ can represent different operations, depending on what the rows and columns of A mean. The next activities show three common patterns: scoring, selecting, and measuring changes.

Activity: Scoring objects by features

Let the rows of X represent three objects with two features:

$$X = \begin{bmatrix} 3 & 1 \\ 1 & 4 \\ 2 & 2 \end{bmatrix}, \qquad \mathbf{w} = (2, -1).$$

Compute $X\mathbf{w}$. Interpret the result.

We have

$$X\mathbf{w} = \begin{bmatrix} 3 & 1 \\ 1 & 4 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 5 \\ -2 \\ 2 \end{bmatrix}.$$

Each output is a score:

$$2(\text{first feature}) - (\text{second feature}).$$

Activity: A selector matrix

Let

$$S = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \qquad \mathbf{x} = \begin{bmatrix} 5 \\ 6 \\ 7 \\ 8 \end{bmatrix}.$$

Compute $S\mathbf{x}$. What does S do?

We have

$$S\mathbf{x} = \begin{bmatrix} 1\cdot 5 + 0\cdot 6 + 0\cdot 7 + 0\cdot 8 \\ 0\cdot 5 + 0\cdot 6 + 1\cdot 7 + 0\cdot 8 \end{bmatrix} = \begin{bmatrix} 5 \\ 7 \end{bmatrix}.$$

The matrix S selects the first and third entries of $\mathbf{x}$.

Activity: A difference matrix

Let

$$D = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}, \qquad \mathbf{x} = \begin{bmatrix} 2 \\ 5 \\ 9 \end{bmatrix}.$$

Compute $D\mathbf{x}$. What does D measure?

We have

$$
\begin{aligned}
D\mathbf{x} &= \begin{bmatrix} -1 \cdot 2 + 1 \cdot 5 + 0 \cdot 9 \\ 0 \cdot 2 + (-1) \cdot 5 + 1 \cdot 9 \end{bmatrix} \\
&= \begin{bmatrix} 3 \\ 4 \end{bmatrix}.
\end{aligned}
$$

The entries are consecutive differences:

$$5 - 2 = 3, \qquad 9 - 5 = 4.$$

Thus D measures changes in a short time series.

Activity: Document ranking in matrix code

In Word-count vectors, we ranked documents by cosine similarity one score at a time. The following code repeats that ranking using a data matrix whose rows are the document vectors.

```
import numpy as np

doc_names = np.array(["D1", "D2", "D3"])

X = np.array([
    [2, 0, 2],
    [1, 1, 0],
    [0, 2, 0],
], dtype=float)

q = np.array([1, 0, 1], dtype=float)

scores = (X @ q) / (np.linalg.norm(X, axis=1) * np.linalg.norm(q))
ranking = doc_names[np.argsort(scores)[::-1]]
```

Activity: Document ranking in matrix code (continued)

```
scores, ranking
```

Output:

```
(array([1. , 0.5, 0. ]),
 array(['D1', 'D2', 'D3'], dtype='<U2'))
```

1. Which row of `X` is the vector $\mathbf{D}_2$?
2. Which expression computes the three dot products $\mathbf{D}_i \cdot \mathbf{q}$?
3. What does `np.argsort(scores)[::-1]` do?
4. Why does the output agree with Word-count vectors?

The rows of `X` are $\mathbf{D}_1$, $\mathbf{D}_2$, and $\mathbf{D}_3$. The product `X @ q` computes the three dot products with the query. The expression `np.argsort(scores)[::-1]` gives the indices of the scores from largest to smallest. The output agrees with Word-count vectors: $1, \frac{1}{2}, 0$.

Example: The attention activity in matrix form

In A tiny attention-style token-weighting calculation, we computed the scores

$$\mathbf{q} \cdot \mathbf{k}_{\text{small}}, \qquad \mathbf{q} \cdot \mathbf{k}_{\text{red}}, \qquad \mathbf{q} \cdot \mathbf{k}_{\text{bird}}$$

one at a time. Matrix-vector multiplication computes the same scores at once.

Put the key vectors as the rows of

$$K = \begin{bmatrix} \mathbf{k}_{\text{small}}^T \\ \mathbf{k}_{\text{red}}^T \\ \mathbf{k}_{\text{bird}}^T \end{bmatrix}.$$

Then

$$K\mathbf{q} = \begin{bmatrix} \mathbf{k}_{\text{small}} \cdot \mathbf{q} \\ \mathbf{k}_{\text{red}} \cdot \mathbf{q} \\ \mathbf{k}_{\text{bird}} \cdot \mathbf{q} \end{bmatrix}.$$

For the numbers in A tiny attention-style token-weighting calculation,

$$K = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}, \qquad \mathbf{q} = \begin{bmatrix} 1 \\ 1 \end{bmatrix},$$

so

$$K\mathbf{q} = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}.$$

Activity: The same attention calculation in code

The following code performs the calculation from the token attention activity.

```
import numpy as np

tokens = np.array(["small", "red", "bird"])

q = np.array([1.0, 1.0])
```

Activity: The same attention calculation in code (continued)

```
K = np.array([
    [1.0, 0.0],
    [0.0, 1.0],
    [1.0, 1.0],
])

V = np.array([
    [4.0, 0.0],
    [0.0, 4.0],
    [4.0, 4.0],
])

s = K @ q
alpha = s / s.sum()
out = alpha @ V

s, alpha, out
```

Output:

```
(array([1., 1., 2.]),
 array([0.25, 0.25, 0.5 ]),
 array([3., 3.]))
```

1. Which line computes the token scores?
2. Which token receives the largest weight?
3. Which line forms the weighted average?
4. Why do the entries of `alpha` add to 1?

The line `s = K @ q` computes the three dot products. The token “bird” receives the largest weight. The line `out = alpha @ V` forms the weighted average of the rows of `V`. The entries of `alpha` add to 1 because the scores were divided by their sum.

Geometric matrix actions in $\mathbb{R}^2$

Definition: Identity matrix

The identity matrix I_n is the $n \times n$ matrix with 1s on the main diagonal and zeros elsewhere.

Fact

For every column vector $\mathbf{x} \in \mathbb{R}^n$, $I_n \mathbf{x} = \mathbf{x}$.

Definition: Standard basis

The j-th standard basis vector e_j is the column vector with a 1 in position j and zeros elsewhere.

Fact

If $A = \begin{bmatrix} \mathbf{a}_1 & \cdots & \mathbf{a}_n \end{bmatrix}$, then $A\mathbf{e}_j = \mathbf{a}_j$. Applying a matrix to a standard basis vector picks out a column.

In $\mathbb{R}^2$, a matrix $A = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 \end{bmatrix}$ sends $\mathbf{e}_1$ to $\mathbf{a}_1$ and $\mathbf{e}_2$ to $\mathbf{a}_2$. Since every vector $\mathbf{x} = (x_1, x_2)$ can be written as $x_1\mathbf{e}_1 + x_2\mathbf{e}_2$, we have

$$A\mathbf{x} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2.$$

Thus a 2×2 matrix is geometrically determined by where it sends the two coordinate directions.

To understand a 2×2 matrix A, apply it to the four corners of the unit square. The image is usually a parallelogram. The two edges leaving the origin are the columns of A. We will refer to this as the

$\boxed{\text{unit-square visualization}}$.

- **Matrix:** $\begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$
  Ae_1: $\begin{bmatrix} 2 \\ 0 \end{bmatrix}$
  Ae_2: $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$
  Name: horizontal stretch
  Unit-square effect: The right edge moves farther right.

- **Matrix:** $\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$
  Ae_1: $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$
  Ae_2: $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$
  Name: projection onto the y-axis
  Unit-square effect: Horizontal information is forgotten.

- **Matrix:** $\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$
  Ae_1: $\begin{bmatrix} -1 \\ 0 \end{bmatrix}$

Ae_2: $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$
Name: reflection across the y-axis
Unit-square effect: The square flips left.

- **Matrix:** $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
  Ae_1: $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$
  Ae_2: $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$
  Name: reflection across the line $y = x$
  Unit-square effect: The square stays in the same region, but the horizontal and vertical coordinate directions exchange.

- **Matrix:** $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$
  Ae_1: $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$
  Ae_2: $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$
  Name: horizontal shear
  Unit-square effect: The bottom edge stays fixed; the top edge shifts right.

- **Matrix:** $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$
  Ae_1: $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$
  Ae_2: $\begin{bmatrix} -1 \\ 0 \end{bmatrix}$
  Name: 90° counterclockwise rotation
  Unit-square effect: The square rotates around the origin.

- **Matrix:** $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$

$A\mathbf{e}_1$: $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$
$A\mathbf{e}_2$: $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$
Name: projection onto the x-axis
Unit-square effect: Vertical information is forgotten.

For more illustrations of these transformations, see Lab U1: Vectors, Similarity, Attention, and Matrix Actions.

Example: Which column tells the story?

For each matrix in the compact transformation gallery, compare $A\mathbf{e}_1$ and $A\mathbf{e}_2$ with $\mathbf{e}_1$ and $\mathbf{e}_2$. Which column changed most visibly from the identity matrix? What does that change do to the unit square?

The columns $A\mathbf{e}_1$ and $A\mathbf{e}_2$ are the transformed coordinate directions. A stretch changes the length of a column. An axis reflection reverses one coordinate direction. Reflection across $y = x$ exchanges the two coordinate directions. A shear changes one column by adding part of the other direction. A projection sends one column to zero, so one direction is forgotten.

Activity: Reflection across the y-axis

Let $A = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$. Compute $A\mathbf{e}_1$, $A\mathbf{e}_2$, $A\mathbf{x}$, and $A\mathbf{y}$, where $\mathbf{x} = (1, 1)$ and $\mathbf{y} = (0, 1)$. Which coordinate changes?

We get

$$A\mathbf{e}_1 = \begin{bmatrix} -1 \\ 0 \end{bmatrix}, \qquad A\mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix},$$
$$A\mathbf{x} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}, \qquad A\mathbf{y} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

The first coordinate flips sign and the second coordinate is unchanged.

Activity: Reflection across the line y=x

Let

$$M = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

1. Compute $M\mathbf{e}_1$ and $M\mathbf{e}_2$.
2. Compute $M\begin{bmatrix} x \\ y \end{bmatrix}$.
3. Which coordinates are exchanged?

Activity: Reflection across the line y=x (continued)

4. What happens to a point on the line $y = x$?
5. Why can an unlabeled unit-square image appear unchanged?

We have

$$M\mathbf{e}_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \mathbf{e}_2, \qquad M\mathbf{e}_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \mathbf{e}_1.$$

More generally, write the input in terms of the standard basis and use linearity:

$$\begin{aligned} M\begin{bmatrix} x \\ y \end{bmatrix} &= M(x\mathbf{e}_1 + y\mathbf{e}_2) = xM\mathbf{e}_1 + yM\mathbf{e}_2 \\ &= x\mathbf{e}_2 + y\mathbf{e}_1 = \begin{bmatrix} y \\ x \end{bmatrix}. \end{aligned}$$

Thus the two coordinates are exchanged. If $x = y$, exchanging the coordinates does not move the point, so every point on the line $y = x$ stays fixed. The unit square occupies the same region after the reflection, even though its horizontal and vertical coordinate directions have exchanged, so an unlabeled image of the square can appear unchanged.

Activity: Horizontal shear

Let $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$. Compute $A\mathbf{e}_1$, $A\mathbf{e}_2$, and $A\mathbf{x}$, where $\mathbf{x} = (1, 1)$. Explain why the bottom edge stays fixed and the top edge shifts right.

We have

$$A\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad A\mathbf{e}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad A\mathbf{x} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

The formula is $A(x, y) = (x + y, y)$. Points with $y = 0$ stay fixed, and points with $y = 1$ shift right by 1.

Note

For the shear matrix $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, the row view gives

$$A \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} (1, 1) \cdot (x, y) \\ (0, 1) \cdot (x, y) \end{bmatrix} = \begin{bmatrix} x + y \\ y \end{bmatrix}.$$

The first row measures horizontal plus vertical, while the second row measures vertical only. The column view gives

$$A\begin{bmatrix} x \\ y \end{bmatrix} = x\begin{bmatrix} 1 \\ 0 \end{bmatrix} + y\begin{bmatrix} 1 \\ 1 \end{bmatrix},$$

so the y-coordinate contributes both upward and rightward motion.

Linear and affine maps

A function assigns one output to each input. In this course, many functions have vectors as inputs and vectors as outputs:

$$F : \mathbb{R}^n \to \mathbb{R}^m.$$

A matrix gives one important source of such functions by the rule $\mathbf{x} \mapsto A\mathbf{x}$.

Definition: Matrix map

Given an $m \times n$ matrix A, the $\boxed{\text{matrix map}}$ induced by A is the function $T_A : \mathbb{R}^n \to \mathbb{R}^m$ defined by

$$T_A(\mathbf{x}) = A\mathbf{x}.$$

Definition: Linear map

A function $T : \mathbb{R}^n \to \mathbb{R}^m$ is $\boxed{\text{linear}}$ if

$$T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v}),$$
$$T(c\mathbf{v}) = cT(\mathbf{v})$$

for all vectors $\mathbf{u}, \mathbf{v}$ and all scalars c.

Theorem: Matrix maps are linear

If A is an $m \times n$ matrix, then the function

$$T_A : \mathbb{R}^n \to \mathbb{R}^m, \qquad T_A(\mathbf{x}) = A\mathbf{x}$$

is linear.

Proof

Matrix-vector multiplication distributes over vector addition and scalar multiplication:

$$A(\mathbf{u} + \mathbf{v}) = A\mathbf{u} + A\mathbf{v}, \qquad A(c\mathbf{v}) = cA\mathbf{v}.$$

These are exactly the two linearity rules.

Theorem: Every linear map has a matrix

Every linear map $T : \mathbb{R}^n \to \mathbb{R}^m$ is given by multiplication by a unique matrix. The columns of that matrix are $T(\mathbf{e}_1), \dots, T(\mathbf{e}_n)$.

Activity: Images of basis vectors determine the matrix

Suppose a linear map $T : \mathbb{R}^2 \to \mathbb{R}^2$ satisfies $T(\mathbf{e}_1) = (2, 1)$ and $T(\mathbf{e}_2) = (-1, 3)$. Find the matrix A such that $T(\mathbf{x}) = A\mathbf{x}$.

The columns are the images of the standard basis vectors, so

$$A = \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix}.$$

Warning

We use “linear map” and “linear transformation” interchangeably; “transformation” is often used when emphasizing geometry.

Definition: Affine map

A function of the form

$$\mathbf{x} \mapsto A\mathbf{x} + \mathbf{b}$$

is called $\boxed{\text{affine}}$. It is built from a linear map followed by a translation.

An affine map is linear only when $\mathbf{b} = \mathbf{0}$. A quick test is the zero vector: every linear map sends $\mathbf{0}$ to $\mathbf{0}$, but

$$A\mathbf{0} + \mathbf{b} = \mathbf{b}.$$

If $\mathbf{b} \neq \mathbf{0}$, the map is not linear.

Example: An affine layer

A layer in a neural network can take the form

$$\mathbf{x} \mapsto W\mathbf{x} + \mathbf{b}.$$

The matrix W mixes the input coordinates. The vector $\mathbf{b}$ shifts the result. Since

$$W\mathbf{0} + \mathbf{b} = \mathbf{b},$$

this map is affine. It is linear exactly when $\mathbf{b} = \mathbf{0}$.